Kunal Thorvat

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Professional Summary

Staff Software Engineer with 10+ years of experience designing and architecting enterprise-scale, high-performance systems in distributed and cloud environments. Proven record of leading cross-functional teams, driving technical strategy, and delivering resilient SaaS platforms. Recognized for architecture leadership, mentorship, and close collaboration with product and design teams to translate business goals into scalable engineering solutions.

Core Skills

Languages & Frameworks: Java, Spring Boot, ReactJS, Angular, JavaScript
Cloud & Infrastructure / Distributed Systems: GCP, Azure, Kubernetes, Kafka, BigQuery
Architecture & Observability: Microservices, Service Operations, Dynatrace, Prometheus, Performance Engineering
Product & Cross-Functional Delivery: Full-stack (UI, API, Backend, Data); partnering with product and design to align technical direction with business objectives
Leadership & Mentorship: Architecture reviews, code governance, team coaching, unblocking execution, promoting best practices (security, reliability, maintainability)
Other Tools: GitHub Copilot, Windsurf, Confluence, Jira

Professional Experience

Staff Software Engineer | Walmart Inc.

Bentonville, AR | Jan 2018 – Present

Scope & Impact: Technical lead and architect for mission-critical workforce-management platforms supporting 2 M+ associates worldwide. Owned architecture decisions, partnered with product and business teams to shape requirements, mentored developers, and drove cross-team delivery of Tier-0 systems with high reliability and observability.

Scheduling System (Technical Lead & Architect)

- Led architecture and technical direction for Walmart's global Scheduling Platform (Spring Boot, GraphQL, React, Azure SQL), replacing a costly third-party vendor solution.
- Collaborated with product managers, operations, and compliance teams to gather requirements and model complex scheduling rules, labor laws, and availability constraints.
- Designed scalable APIs and data flows that allowed stores to generate and manage schedules directly through store dashboards and mobile apps.
- Introduced reactive architecture and optimized connection management for consistent performance during peak scheduling loads.
- Embedded Dynatrace and Prometheus dashboards to monitor SLA compliance and proactively detect issues.
    - Outcome*:* Transitioned Walmart's workforce scheduling to an in-house, cloud-native platform supporting 2 M+ associates, saving millions in vendor costs, improving scheduling accuracy, and enhancing associate satisfaction through predictable and transparent schedules.

Budget & Demand System (Architectural Owner)

- Architected and led development of a centralized labor-demand and budgeting platform using Apache Airflow DAGs and BigQuery, replacing manual Excel-based workflows for budget planning.
- Partnered with product, finance, and workforce analytics teams to gather requirements, define KPIs, and translate business logic into automated, auditable pipelines.
- Designed scalable data pipelines and APIs that standardized budget inputs and integrated with the Scheduling System to generate data-driven schedules aligned with labor demand.
- Guided engineers on workflow orchestration, data validation, and Airflow best practices for reliable and transparent forecasting runs.

  - Outcome*:* Eliminated spreadsheet-driven budget planning, automated demand forecasting for thousands of stores, and tightly aligned forecasts with associate scheduling — reducing over/under-staffing and improving labor accuracy and efficiency.

Freight Planning Tool (Lead Engineer & System Designer)

- Owned architecture and implementation of a full-stack system (Spring Boot + React) integrating DB2, Informix, BigQuery, and Azure SQL to generate optimal freight unloading assignments.
- Led a team to implement hybrid caching and asynchronous refresh mechanisms for sub-second responses despite legacy sources.
- Defined API boundaries, data contracts, and failover strategies for high availability and fault tolerance. *Outcome:* Improved freight throughput and associate allocation efficiency across hundreds of stores.

Other Major Contributions

- ***Win The Day Incentive System****:* Consolidated 30+ microservices into a modular Spring Boot architecture with Kafka ingestion and Azure SQL persistence; boosted order-filling rates and productivity.
- ***Fixed Scheduling Solution****:* Developed algorithms for quarterly fixed schedules based on availability and compliance rules to enhance predictability and retention.
- ***Deployment & Operations****:* Established Kubernetes-based CI/CD pipelines for Tier-0 systems; ensured zero-downtime deployments and > 80 % test coverage.
- ***Leadership & Mentoring****:* Provided architectural governance across teams, mentored engineers in system design and observability, drove code reviews and best practices.
- ***AI-Assisted Development****:* Championed GitHub Copilot and Windsurf to accelerate delivery and improve PR review quality.

Identity & Access Management Developer | Sath Inc.

Schaumburg, IL | Jun 2016 – Jan 2018

- Automated application onboarding to Oracle Identity Management via MEAN-stack Identity Hub (reduced onboarding from hours to minutes).
- Built User Provisioning Dashboard (Angular + Spring Boot) and custom REST APIs for Oracle OIM reconciliation.
- Developed custom drivers for NetIQ Identity Management and integrated OAuth/SAML SSO for enterprise apps.

Software Engineer | Tech Mahindra Ltd.

Pune, India | Dec 2010 – Jun 2014

- Led team of five to build AT&T’s Template-Based Order Capture Framework (JSF 2.0, Core Java, Servlets, multithreading).
- Developed ISP-e Dashboard for AT&T order tracking (Java/JSP/JSF + SQL); authored HLD/LLD docs and managed Unix deployments.

Education

M.S., Computer Science — University of Illinois at Chicago (2014 – 2016) GPA 3.8 / 4.0
B.E., Information Technology — Mumbai University (2006 – 2010) GPA 3.66 / 4.0